

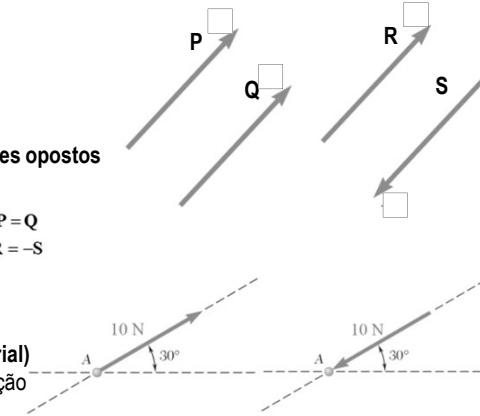
Estática dos Pontos Materiais FORÇAS NO PLANO

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Revisão 2021-II

Grandezas vetoriais

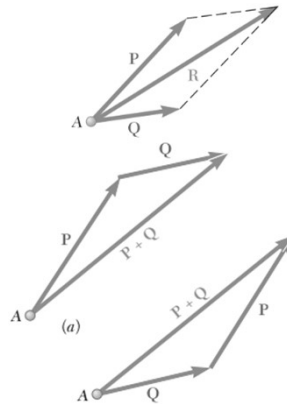
- Vetor**
 - Intensidade
 - Direção
 - Sentido
- Vetores iguais e vetores opostos**
- Notação**

$$\begin{cases} \vec{P} = \vec{Q} \\ \vec{R} = -\vec{S} \end{cases} \quad \text{ou} \quad \begin{cases} \mathbf{P} = \mathbf{Q} \\ \mathbf{R} = -\mathbf{S} \end{cases}$$
- Módulo**
 - $|\mathbf{P}|$ ou $|\vec{P}|$
- Força (grandeza vetorial)**
 - Ponto de aplicação



Operações com vetores

- Soma de vetores**
 - Regra do paralelogramo
$$\mathbf{R} = \mathbf{P} + \mathbf{Q}$$

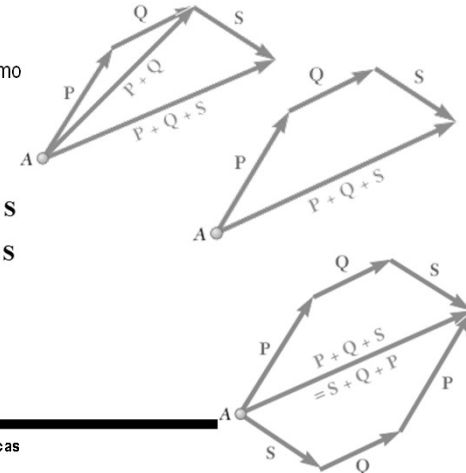


Operações com vetores

- Soma de vetores**
 - Regra do paralelogramo
 - Operação comutativa
 - Operação associativa
$$(\mathbf{P} + \mathbf{Q}) = \mathbf{P} + \mathbf{Q}$$

$$(\mathbf{P} + \mathbf{Q} + \mathbf{S}) = (\mathbf{P} + \mathbf{Q}) + \mathbf{S}$$

$$(\mathbf{P} + \mathbf{Q} + \mathbf{S}) = \mathbf{P} + \mathbf{Q} + \mathbf{S}$$

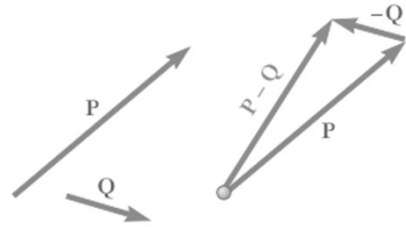


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Operações com vetores

- **Soma de vetores**
 - Regra do paralelogramo
 - Operação comutativa
 - Operação associativa
- **Subtração de vetores**

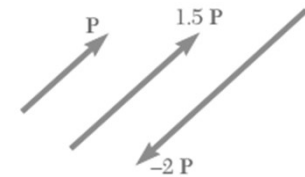
$$\mathbf{P} - \mathbf{Q} = \mathbf{P} + (-\mathbf{Q})$$



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Operações com vetores

- **Soma de vetores**
 - Regra do paralelogramo
 - Operação comutativa
 - Operação associativa
- **Subtração de vetores**
- **Multiplificação de um vetor por um escalar**



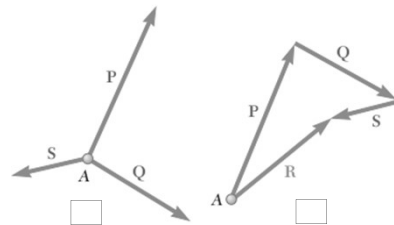
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Forças concorrentes

- **Resultante**

$$\mathbf{R} = \sum \mathbf{F}_i$$

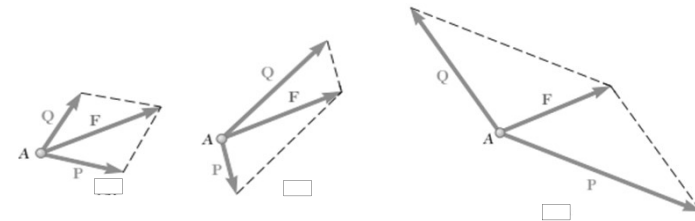
$$\mathbf{R} = \mathbf{P} + \mathbf{Q} + \mathbf{S}$$



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Decomposição de forças

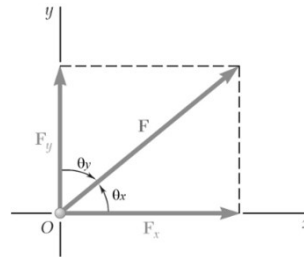
- **Componentes de uma força**



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Decomposição de forças

- Componentes de uma força
- Componentes retangulares de uma força



$$F_x = F \cos \theta_x$$

$$F_y = F \sin \theta_x$$

$$F = \sqrt{F_x^2 + F_y^2}$$

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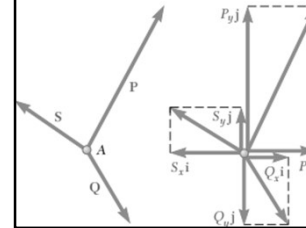
Decomposição de forças

- Componentes de uma força
- Componentes retangulares de uma força
- Soma de vetores por componentes cartesianas

$$\mathbf{R} = \mathbf{P} + \mathbf{Q} + \mathbf{S}$$

$$R_x = \sum F_x \Rightarrow R_x = P_x + Q_x + S_x$$

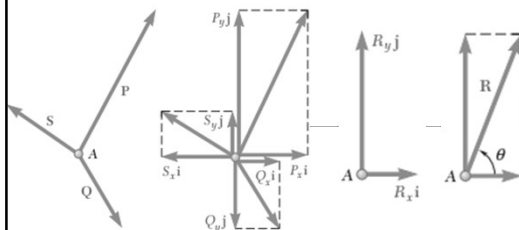
$$R_y = \sum F_y \Rightarrow R_y = P_y + Q_y + S_y$$



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Decomposição de forças

- Componentes de uma força
- Componentes retangulares de uma força
- Soma de vetores por componentes cartesianas
- Vetor resultante de uma soma de vetores



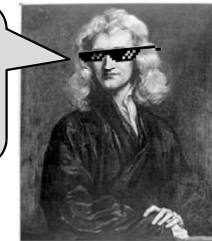
$$R = \sqrt{R_x^2 + R_y^2}$$

$$\tan \theta = \frac{R_y}{R_x}$$

Condição de equilíbrio

- Primeira lei de newton

"Se a força resultante que atua sobre um ponto material tem intensidade igual a zero, esse ponto permanece em repouso (se estava originalmente em repouso) ou se move com velocidade constante (se estava originalmente em movimento)."



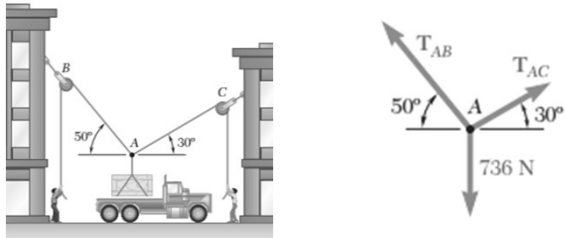
- Equilíbrio de um ponto material no plano

$$\mathbf{R} = \sum \mathbf{F} = \mathbf{0} \Rightarrow \begin{aligned} \sum F_y &= 0 \\ \sum F_x &= 0 \end{aligned}$$

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Diagrama de corpo livre

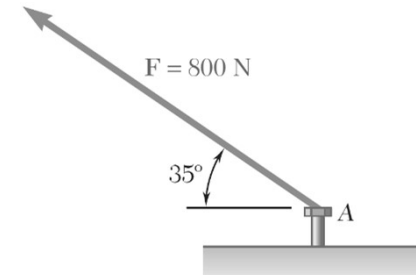
- Identificar a partícula que deve ser equilibrada
- Isolar as partículas dos elementos que interagem com ela
- Identificar e representar as forças que agem na partícula



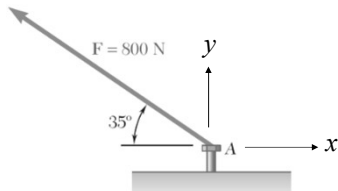
14

Exemplo 1:

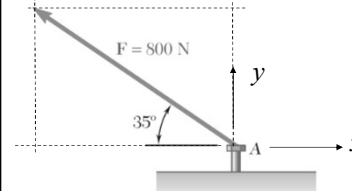
Uma força de 800 N é exercida sobre um parafuso A, como mostrado na figura. Determine as componentes vertical e horizontal.



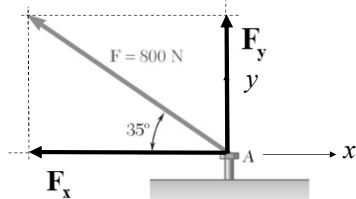
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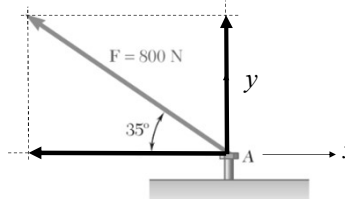
17



$$F_x = F \cos \theta_x$$

$$F_y = F \sin \theta_x$$

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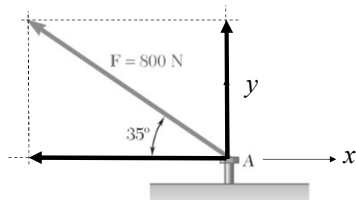
$$F_x = F \cos \theta_x$$

$$F_x = F \cos 35^\circ$$

$$F_y = F \sin \theta_x$$

$$F_y = F \sin 35^\circ$$

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$$F_x = F \cos \theta_x$$

$$F_x = F \cos 35^\circ$$

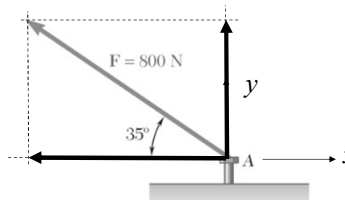
$$F_x = F \cdot 0,819$$

$$F_y = F \sin \theta_x$$

$$F_y = F \sin 35^\circ$$

$$F_y = F \cdot 0,574$$

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$$F_x = F \cos \theta_x$$

$$F_x = F \cos 35^\circ$$

$$F_x = F \cdot 0,819$$

$$F_x = 655,2\text{N} (\leftarrow)$$

$$F_y = F \sin \theta_x$$

$$F_y = F \sin 35^\circ$$

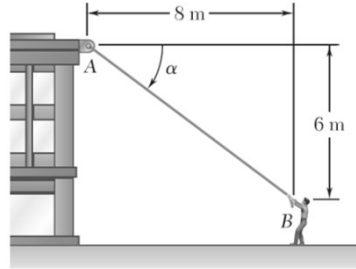
$$F_y = F \cdot 0,574$$

$$F_y = 459,2\text{N} (\uparrow)$$

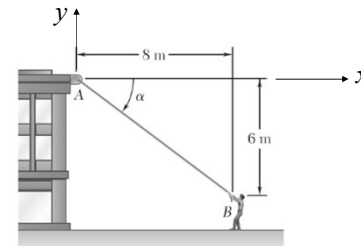
21

Exemplo 2:

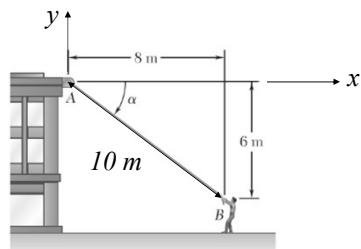
Um homem puxa, com uma força de 300 N, uma corda fixada a uma construção, como mostrado na figura. Determine as componentes vertical e horizontal dessa força.



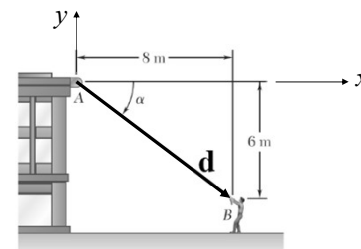
22



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Diagram showing a force F acting on a structure. The force is applied at point A , which is 8 m horizontally and 6 m vertically from point B . The force vector F is directed towards B at an angle α to the horizontal. A dashed line segment d is shown along the line of action of the force.

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Diagram showing a force F acting on a structure. The force is applied at point A , which is 8 m horizontally and 6 m vertically from point B . The force vector F is directed towards B at an angle α to the horizontal. A dashed line segment d is shown along the line of action of the force.

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Diagram showing a force F acting on a structure. The force is applied at point A , which is 8 m horizontally and 6 m vertically from point B . The force vector F is directed towards B at an angle α to the horizontal. Dashed lines indicate the horizontal and vertical components of the force.

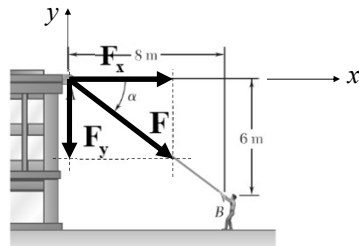
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Diagram showing a force F acting on a structure. The force is applied at point A , which is 8 m horizontally and 6 m vertically from point B . The force vector F is directed towards B at an angle α to the horizontal. The horizontal component is labeled F_x and the vertical component is labeled F_y .

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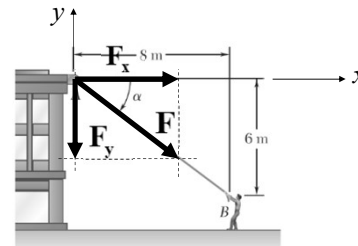
29



$$F_x = F \cos \theta_x$$

$$F_y = F \sin \theta_x$$

30



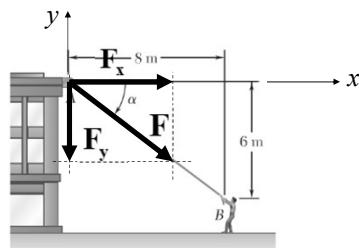
$$F_x = F \cos \theta_x$$

$$F_x = F \cdot \frac{8}{10}$$

$$F_y = F \sin \theta_x$$

$$F_y = F \cdot \frac{6}{10}$$

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$$F_x = F \cos \theta_x$$

$$F_x = F \cdot \frac{8}{10}$$

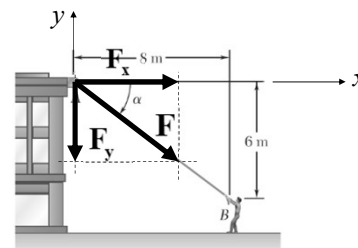
$$F_x = F \cdot 0,8$$

$$F_y = F \sin \theta_x$$

$$F_y = F \cdot \frac{6}{10}$$

$$F_y = F \cdot 0,6$$

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$$F_x = F \cos \theta_x$$

$$F_x = F \cdot \frac{8}{10}$$

$$F_x = F \cdot 0,8$$

$$F_x = 240\text{N} (\rightarrow)$$

$$F_y = F \sin \theta_x$$

$$F_y = F \cdot \frac{6}{10}$$

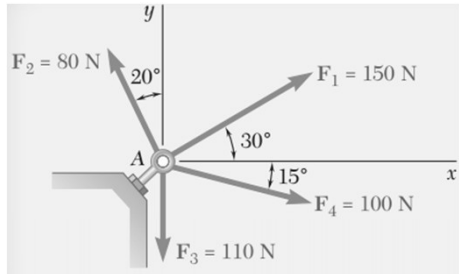
$$F_y = F \cdot 0,6$$

$$F_y = 180\text{N} (\downarrow)$$

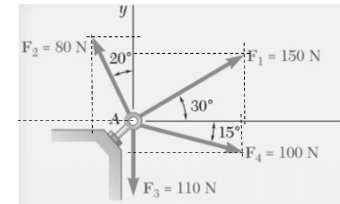
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Exemplo 3:

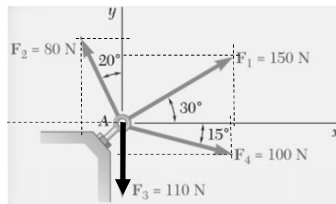
Determine a resultante de forças no parafuso A:



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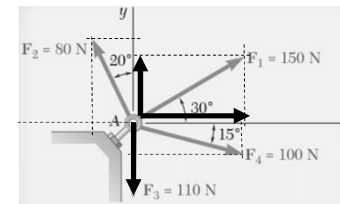
35



$$F_{3x} = 0$$

$$F_{3y} = -110\text{ N}$$

36



$$F_{3x} = 0$$

$$F_{3y} = -110\text{ N}$$

$$F_{1x} = F_1 \cos 30^\circ$$

$$F_{1y} = F_1 \sin 30^\circ$$

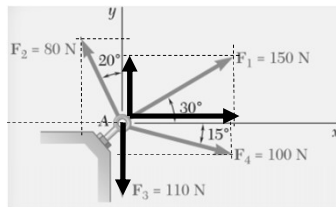
$$F_{1x} = 150 \cdot 0,866$$

$$F_{1y} = 150 \cdot 0,5$$

$$F_{1x} = 129,9\text{ N}$$

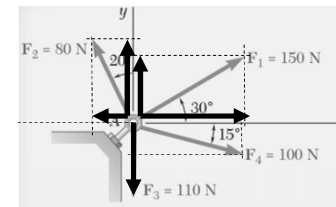
$$F_{1y} = 75\text{ N}$$

37



$$\begin{aligned} F_{3x} &= 0 & F_{3y} &= -110\text{N} \\ F_{1x} &= 129,9\text{N} & F_{1y} &= 75\text{N} \end{aligned}$$

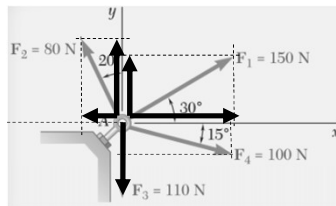
38



$$\begin{aligned} F_{3x} &= 0 & F_{3y} &= -110\text{N} \\ F_{1x} &= 129,9\text{N} & F_{1y} &= 75\text{N} \end{aligned}$$

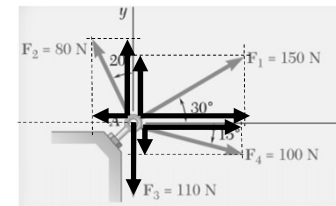
$$\begin{aligned} F_{2x} &= -F_2 \cdot \sin 20^\circ & F_{2y} &= F_2 \cdot \cos 20^\circ \\ F_{2x} &= -80 \cdot 0,342 & F_{2y} &= 80 \cdot 0,940 \\ F_{2x} &= -27,4\text{N} & F_{2y} &= 75,2\text{N} \end{aligned}$$

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$$\begin{aligned} F_{3x} &= 0 & F_{3y} &= -110\text{N} \\ F_{1x} &= 129,9\text{N} & F_{1y} &= 75\text{N} \\ F_{2x} &= -27,4\text{N} & F_{2y} &= 75,2\text{N} \end{aligned}$$

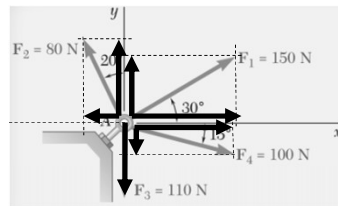
40



$$\begin{aligned} F_{3x} &= 0 & F_{3y} &= -110\text{N} \\ F_{1x} &= 129,9\text{N} & F_{1y} &= 75\text{N} \\ F_{2x} &= -27,4\text{N} & F_{2y} &= 75,2\text{N} \end{aligned}$$

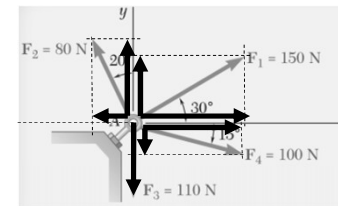
$$\begin{aligned} F_{4x} &= F_4 \cdot \cos 15^\circ & F_{4y} &= -F_4 \cdot \sin 15^\circ \\ F_{4x} &= 100 \cdot 0,966 & F_{4y} &= -100 \cdot 0,259 \\ F_{4x} &= 96,6\text{N} & F_{4y} &= -25,9\text{N} \end{aligned}$$

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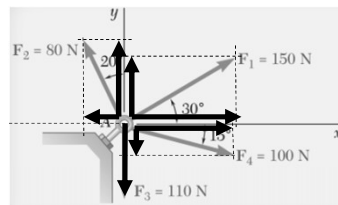
$$\begin{array}{ll}
 F_{3x} = 0 & F_{3y} = -110\text{ N} \\
 F_{1x} = 129,9\text{ N} & F_{1y} = 75\text{ N} \\
 F_{2x} = -27,4\text{ N} & F_{2y} = 75,2\text{ N} \\
 F_{4x} = 96,6\text{ N} & F_{4y} = -25,9\text{ N}
 \end{array}$$

42



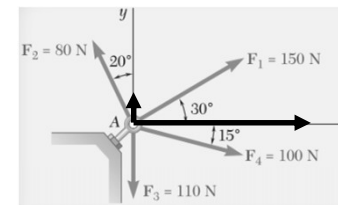
$$\begin{array}{ll}
 F_{3x} = 0 & F_{3y} = -110\text{ N} \\
 F_{1x} = 129,9\text{ N} & F_{1y} = 75\text{ N} \\
 + F_{2x} = -27,4\text{ N} & + F_{2y} = 75,2\text{ N} \\
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 \end{array}$$

43



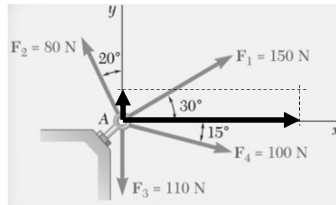
$$\begin{array}{ll}
 F_{3x} = 0 & F_{3y} = -110\text{ N} \\
 F_{1x} = 129,9\text{ N} & F_{1y} = 75\text{ N} \\
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 F_{4x} = 96,6\text{ N} & F_{4y} = -25,9\text{ N} \\
 \hline
 R_x = 199,1\text{ N} & R_y = 14,6\text{ N}
 \end{array}$$

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$$\begin{array}{ll}
 F_{3x} = 0 & F_{3y} = -110\text{ N} \\
 F_{1x} = 129,9\text{ N} & F_{1y} = 75\text{ N} \\
 F_{2x} = -27,4\text{ N} & F_{2y} = 75,2\text{ N} \\
 F_{4x} = 96,6\text{ N} & F_{4y} = -25,9\text{ N} \\
 \hline
 R_x = 199,1\text{ N} (\rightarrow) & \\
 R_y = 14,6\text{ N} (\uparrow) &
 \end{array}$$

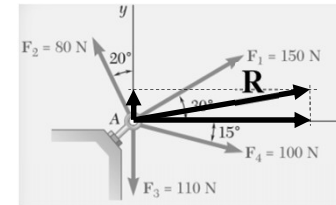
45



$$\begin{aligned} F_{3x} &= 0 & F_{3y} &= -110\text{ N} \\ F_{1x} &= 129,9\text{ N} & F_{1y} &= 75\text{ N} \\ F_{2x} &= -27,4\text{ N} & F_{2y} &= 75,2\text{ N} \\ F_{4x} &= 96,6\text{ N} & F_{4y} &= -25,9\text{ N} \end{aligned}$$

$$\begin{aligned} R_x &= 199,1\text{ N} (\rightarrow) \\ R_y &= 14,6\text{ N} (\uparrow) \end{aligned}$$

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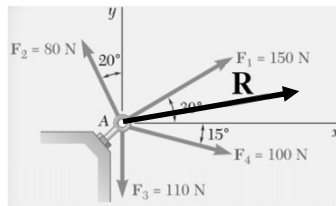


$$\begin{aligned} F_{3x} &= 0 & F_{3y} &= -110\text{ N} \\ F_{1x} &= 129,9\text{ N} & F_{1y} &= 75\text{ N} \\ F_{2x} &= -27,4\text{ N} & F_{2y} &= 75,2\text{ N} \\ F_{4x} &= 96,6\text{ N} & F_{4y} &= -25,9\text{ N} \end{aligned}$$

$$\begin{aligned} R_x &= 199,1\text{ N} (\rightarrow) \\ R_y &= 14,6\text{ N} (\uparrow) \end{aligned}$$

$$R = \sqrt{R_x^2 + R_y^2}$$

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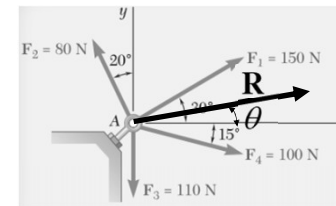


$$\begin{aligned} F_{3x} &= 0 & F_{3y} &= -110\text{ N} \\ F_{1x} &= 129,9\text{ N} & F_{1y} &= 75\text{ N} \\ F_{2x} &= -27,4\text{ N} & F_{2y} &= 75,2\text{ N} \\ F_{4x} &= 96,6\text{ N} & F_{4y} &= -25,9\text{ N} \end{aligned}$$

$$\begin{aligned} R_x &= 199,1\text{ N} (\rightarrow) \\ R_y &= 14,6\text{ N} (\uparrow) \end{aligned}$$

$$\begin{aligned} R &= \sqrt{199,1^2 + 14,6^2} \\ R &= 199,6\text{ N} \end{aligned}$$

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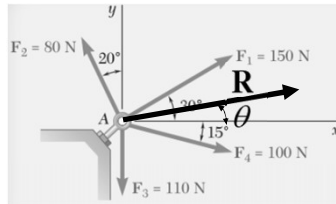
$$\begin{aligned} F_{3x} &= 0 & F_{3y} &= -110\text{ N} \\ F_{1x} &= 129,9\text{ N} & F_{1y} &= 75\text{ N} \\ F_{2x} &= -27,4\text{ N} & F_{2y} &= 75,2\text{ N} \\ F_{4x} &= 96,6\text{ N} & F_{4y} &= -25,9\text{ N} \end{aligned}$$

$$\begin{aligned} R_x &= 199,1\text{ N} (\rightarrow) \\ R_y &= 14,6\text{ N} (\uparrow) \end{aligned}$$

$$R = 199,6\text{ N}$$

$$\tan \theta = \frac{R_y}{R_x}$$

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$$\begin{aligned} F_{3x} &= 0 & F_{3y} &= -110\text{ N} \\ F_{1x} &= 129,9\text{ N} & F_{1y} &= 75\text{ N} \\ F_{2x} &= -27,4\text{ N} & F_{2y} &= 75,2\text{ N} \\ F_{4x} &= 96,6\text{ N} & F_{4y} &= -25,9\text{ N} \end{aligned}$$

$$R_x = 199,1\text{ N} (\rightarrow)$$

$$R_y = 14,6\text{ N} (\uparrow)$$

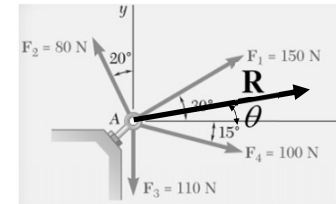
$$R = 199,6\text{ N}$$

$$\theta = \arctan \frac{14,6}{199,1} = 4,2^\circ$$

$$\tan \theta = \frac{R_y}{R_x}$$

UFV
Capítulo 2 - Propriedades Geométricas das Áreas Planas

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$$\begin{aligned} F_{3x} &= 0 & F_{3y} &= -110\text{ N} \\ F_{1x} &= 129,9\text{ N} & F_{1y} &= 75\text{ N} \\ F_{2x} &= -27,4\text{ N} & F_{2y} &= 75,2\text{ N} \\ F_{4x} &= 96,6\text{ N} & F_{4y} &= -25,9\text{ N} \end{aligned}$$

$$R_x = 199,1\text{ N} (\rightarrow)$$

$$R_y = 14,6\text{ N} (\uparrow)$$

$$R = 199,6\text{ N}$$

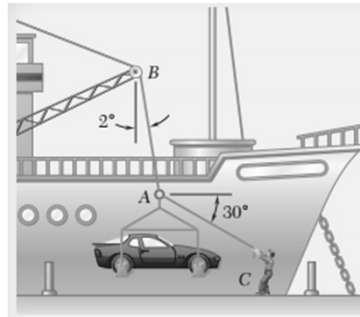
$$\theta = 4,2^\circ$$

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Exemplo 4:

Numa operação de descarga de navio uma automóvel com peso próprio de 17,50 kN é suportado por um cabo. Uma corda amarrada ao cabo em A é puxada a fim de que o automóvel seja centralizado na posição desejada. O ângulo entre o cabo e a vertical é de 2° , enquanto o ângulo entre a corda e a horizontal é de 30° . Qual a força de tração nesta corda?

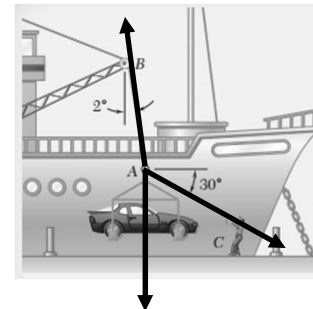


Respostas:

$$T_{AB} = 17,87\text{ kN}; T_{AC} = 0,72\text{ kN}.$$

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UFV
Capítulo 2 - Propriedades Geométricas das Áreas Planas

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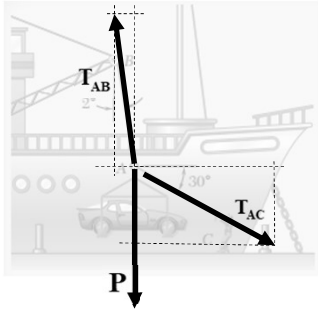


Diagram showing a ship's mast with forces T_{AB} (at 2°), T_{AC} (at 30°), and a vertical force P .



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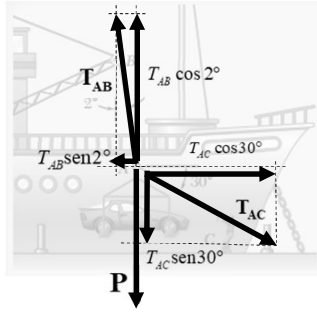


Diagram showing the components of forces T_{AB} and T_{AC} relative to a vertical axis. T_{AB} has components $T_{AB} \cos 2^\circ$ (vertical) and $T_{AB} \sin 2^\circ$ (horizontal). T_{AC} has components $T_{AC} \cos 30^\circ$ (horizontal) and $T_{AC} \sin 30^\circ$ (vertical). A vertical force P is also shown.



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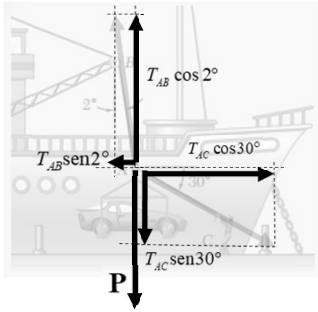


Diagram showing the components of forces T_{AB} and T_{AC} relative to a vertical axis. T_{AB} has components $T_{AB} \cos 2^\circ$ (vertical) and $T_{AB} \sin 2^\circ$ (horizontal). T_{AC} has components $T_{AC} \cos 30^\circ$ (horizontal) and $T_{AC} \sin 30^\circ$ (vertical). A vertical force P is also shown.



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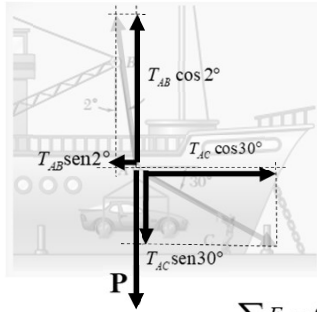


Diagram showing the components of forces T_{AB} and T_{AC} relative to a vertical axis. T_{AB} has components $T_{AB} \cos 2^\circ$ (vertical) and $T_{AB} \sin 2^\circ$ (horizontal). T_{AC} has components $T_{AC} \cos 30^\circ$ (horizontal) and $T_{AC} \sin 30^\circ$ (vertical). A vertical force P is also shown.

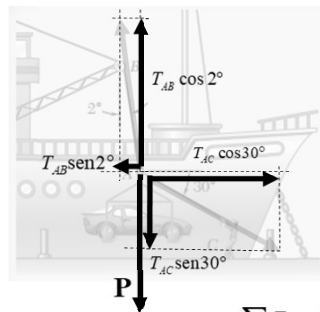


$$\mathbf{R} = \sum \mathbf{F} = \mathbf{0} \Rightarrow \begin{cases} \sum F_x = 0 \\ \sum F_y = 0 \end{cases}$$



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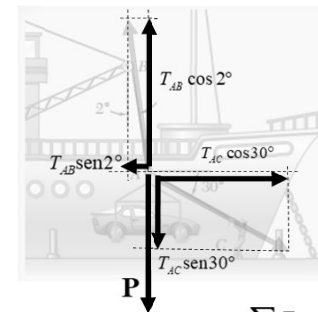
$$\sum F_x = 0$$



$$\mathbf{R} = \sum \mathbf{F} = \mathbf{0} \Rightarrow \begin{aligned} \sum F_x &= 0 \\ \sum F_y &= 0 \end{aligned}$$

Capítulo 2 - Propriedades Geométricas
de Áreas Planas

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$$\sum F_x = 0$$

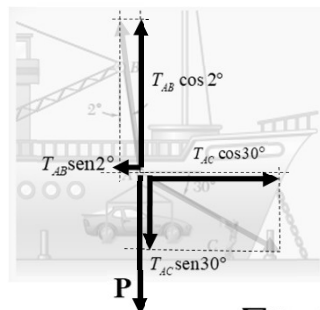
$$T_{AC} \cos 30^\circ - T_{AB} \sin 2^\circ = 0$$



$$\mathbf{R} = \sum \mathbf{F} = \mathbf{0} \Rightarrow \begin{aligned} \sum F_x &= 0 \\ \sum F_y &= 0 \end{aligned}$$

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$$\sum F_x = 0$$

$$T_{AC} \cos 30^\circ - T_{AB} \sin 2^\circ = 0$$

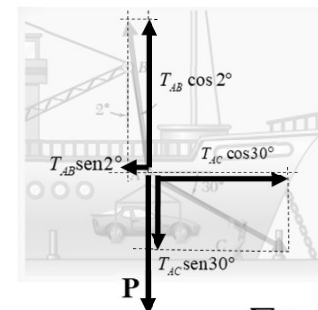
$$T_{AC} = T_{AB} \frac{\sin 2^\circ}{\cos 30^\circ}$$



$$\mathbf{R} = \sum \mathbf{F} = \mathbf{0} \Rightarrow \begin{aligned} \sum F_x &= 0 \\ \sum F_y &= 0 \end{aligned}$$

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de Áreas Planas

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$$\sum F_x = 0$$

$$T_{AC} \cos 30^\circ - T_{AB} \sin 2^\circ = 0$$

$$T_{AC} = T_{AB} \frac{\sin 2^\circ}{\cos 30^\circ}$$

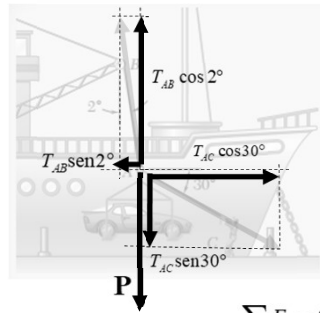
$$\sum F_y = 0$$



$$\mathbf{R} = \sum \mathbf{F} = \mathbf{0} \Rightarrow \begin{aligned} \sum F_x &= 0 \\ \sum F_y &= 0 \end{aligned}$$

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$$\mathbf{R} = \sum \mathbf{F} = \mathbf{0} \Rightarrow \begin{aligned} \sum F_x &= 0 \\ \sum F_y &= 0 \end{aligned}$$

$$\sum F_x = 0$$

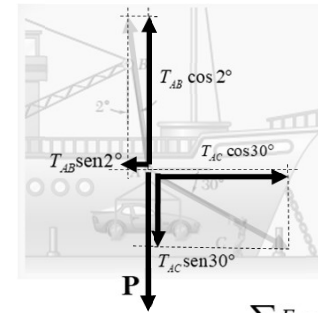
$$T_{AC} \cos 30^\circ - T_{AB} \sin 2^\circ = 0$$

$$T_{AC} = T_{AB} \frac{\sin 2^\circ}{\cos 30^\circ}$$

$$\sum F_y = 0$$

$$T_{AB} \cos 2^\circ - T_{AC} \sin 30^\circ - P = 0$$

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$$\mathbf{R} = \sum \mathbf{F} = \mathbf{0} \Rightarrow \begin{aligned} \sum F_x &= 0 \\ \sum F_y &= 0 \end{aligned}$$

$$\sum F_x = 0$$

$$T_{AC} \cos 30^\circ - T_{AB} \sin 2^\circ = 0$$

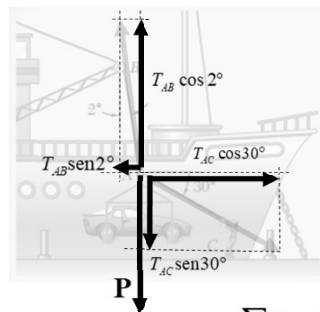
$$T_{AC} = T_{AB} \frac{\sin 2^\circ}{\cos 30^\circ}$$

$$\sum F_y = 0$$

$$T_{AB} \cos 2^\circ - T_{AC} \sin 30^\circ - P = 0$$

$$T_{AB} \cos 2^\circ - T_{AB} \frac{\sin 2^\circ}{\cos 30^\circ} \sin 30^\circ - P = 0$$

63



$$\mathbf{R} = \sum \mathbf{F} = \mathbf{0} \Rightarrow \begin{aligned} \sum F_x &= 0 \\ \sum F_y &= 0 \end{aligned}$$

$$\sum F_x = 0$$

$$T_{AC} \cos 30^\circ - T_{AB} \sin 2^\circ = 0$$

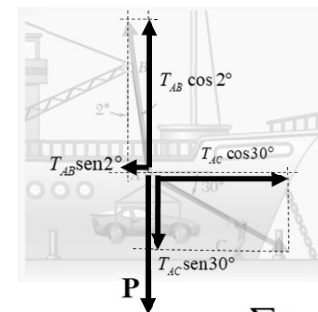
$$T_{AC} = T_{AB} \frac{\sin 2^\circ}{\cos 30^\circ}$$

$$\sum F_y = 0$$

$$T_{AB} \cos 2^\circ - T_{AC} \sin 30^\circ - P = 0$$

$$T_{AB} \cos 2^\circ - T_{AB} \frac{\sin 2^\circ}{\cos 30^\circ} \sin 30^\circ - P = 0$$

64



$$\mathbf{R} = \sum \mathbf{F} = \mathbf{0} \Rightarrow \begin{aligned} \sum F_x &= 0 \\ \sum F_y &= 0 \end{aligned}$$

$$\sum F_x = 0$$

$$T_{AC} \cos 30^\circ - T_{AB} \sin 2^\circ = 0$$

$$T_{AC} = T_{AB} \frac{\sin 2^\circ}{\cos 30^\circ}$$

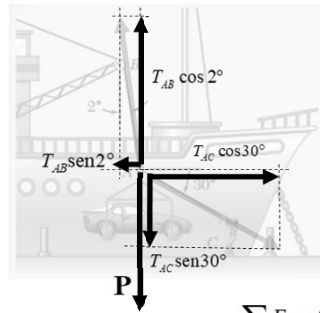
$$\sum F_y = 0$$

$$T_{AB} \cos 2^\circ - T_{AC} \sin 30^\circ - P = 0$$

$$T_{AB} \cos 2^\circ - T_{AB} \frac{\sin 2^\circ}{\cos 30^\circ} \sin 30^\circ - P = 0$$

$$T_{AB} \left(\cos 2^\circ - \frac{\sin 2^\circ}{\cos 30^\circ} \sin 30^\circ \right) - P = 0$$

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$$\mathbf{R} = \sum \mathbf{F} = \mathbf{0} \Rightarrow \begin{aligned} \sum F_x &= 0 \\ \sum F_y &= 0 \end{aligned}$$

$$\sum F_x = 0$$

$$T_{AC} \cos 30^\circ - T_{AB} \sin 2^\circ = 0$$

$$T_{AC} = T_{AB} \frac{\sin 2^\circ}{\cos 30^\circ}$$

$$\sum F_y = 0$$

$$T_{AB} \cos 2^\circ - T_{AC} \sin 30^\circ - P = 0$$

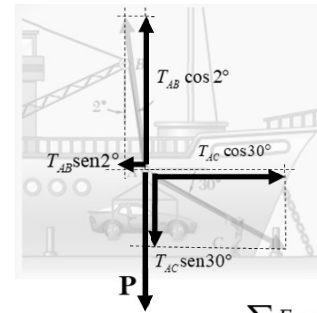
$$T_{AB} \cos 2^\circ - T_{AB} \frac{\sin 2^\circ}{\cos 30^\circ} \sin 30^\circ - P = 0$$

$$T_{AB} \left(\cos 2^\circ - \frac{\sin 2^\circ}{\cos 30^\circ} \sin 30^\circ \right) - P = 0$$

$$T_{AB} = \frac{P}{\left(\cos 2^\circ - \frac{\sin 2^\circ}{\cos 30^\circ} \sin 30^\circ \right)}$$

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$$\mathbf{R} = \sum \mathbf{F} = \mathbf{0} \Rightarrow \begin{aligned} \sum F_x &= 0 \\ \sum F_y &= 0 \end{aligned}$$

$$\sum F_x = 0$$

$$T_{AC} \cos 30^\circ - T_{AB} \sin 2^\circ = 0$$

$$T_{AC} = T_{AB} \frac{\sin 2^\circ}{\cos 30^\circ}$$

$$\sum F_y = 0$$

$$T_{AB} \cos 2^\circ - T_{AC} \sin 30^\circ - P = 0$$

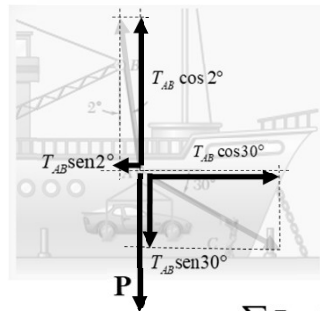
$$T_{AB} \cos 2^\circ - T_{AB} \frac{\sin 2^\circ}{\cos 30^\circ} \sin 30^\circ - P = 0$$

$$T_{AB} \left(\cos 2^\circ - \frac{\sin 2^\circ}{\cos 30^\circ} \sin 30^\circ \right) - P = 0$$

$$T_{AB} = \frac{P}{\left(\cos 2^\circ - \frac{\sin 2^\circ}{\cos 30^\circ} \sin 30^\circ \right)}$$

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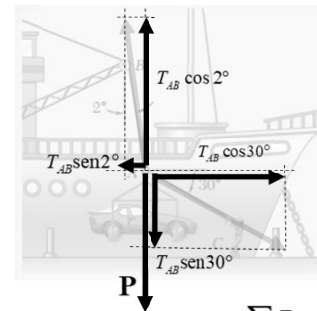
$$\mathbf{R} = \sum \mathbf{F} = \mathbf{0} \Rightarrow \begin{aligned} \sum F_x &= 0 \\ \sum F_y &= 0 \end{aligned}$$

$$T_{AC} = T_{AB} \frac{\sin 2^\circ}{\cos 30^\circ}$$

$$T_{AB} = \frac{P}{\left(\cos 2^\circ - \frac{\sin 2^\circ}{\cos 30^\circ} \sin 30^\circ \right)}$$

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$$\mathbf{R} = \sum \mathbf{F} = \mathbf{0} \Rightarrow \begin{aligned} \sum F_x &= 0 \\ \sum F_y &= 0 \end{aligned}$$

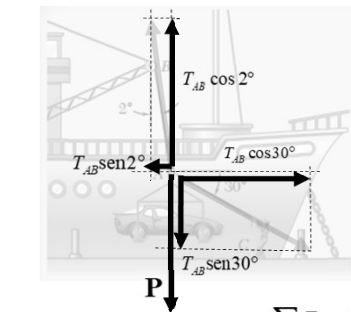
$$T_{AC} = T_{AB} \frac{\sin 2^\circ}{\cos 30^\circ}$$

$$T_{AB} = \frac{P}{\left(\cos 2^\circ - \frac{\sin 2^\circ}{\cos 30^\circ} \sin 30^\circ \right)}$$

$$T_{AB} = \frac{17,5 \text{ kN}}{\left(0,999 - \frac{0,0349}{0,866} 0,5 \right)}$$

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$$\mathbf{R} = \sum \mathbf{F} = \mathbf{0} \Rightarrow \begin{aligned} \sum F_x &= 0 \\ \sum F_y &= 0 \end{aligned}$$

$$T_{AC} = T_{AB} \frac{\sin 2^\circ}{\cos 30^\circ}$$

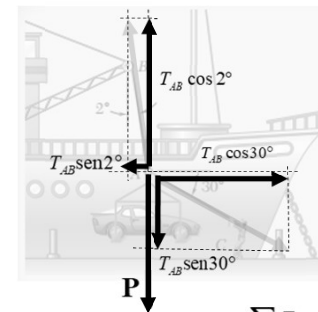
$$T_{AB} = \frac{P}{\left(\cos 2^\circ - \frac{\sin 2^\circ}{\cos 30^\circ} \sin 30^\circ \right)}$$

$$T_{AB} = \frac{17,5 \text{ kN}}{\left(0,999 - \frac{0,0349}{0,866} 0,5 \right)}$$

$$T_{AB} = 17,88 \text{ kN}$$

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$$\mathbf{R} = \sum \mathbf{F} = \mathbf{0} \Rightarrow \begin{aligned} \sum F_x &= 0 \\ \sum F_y &= 0 \end{aligned}$$

$$T_{AC} = T_{AB} \frac{\sin 2^\circ}{\cos 30^\circ}$$

$$T_{AC} = 17,88 \frac{0,0349}{0,866}$$

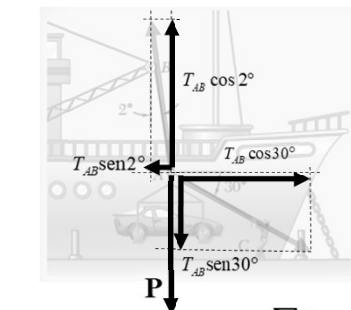
$$T_{AB} = \frac{P}{\left(\cos 2^\circ - \frac{\sin 2^\circ}{\cos 30^\circ} \sin 30^\circ \right)}$$

$$T_{AB} = \frac{17,5 \text{ kN}}{\left(0,999 - \frac{0,0349}{0,866} 0,5 \right)}$$

$$T_{AB} = 17,88 \text{ kN}$$

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$$\mathbf{R} = \sum \mathbf{F} = \mathbf{0} \Rightarrow \begin{aligned} \sum F_x &= 0 \\ \sum F_y &= 0 \end{aligned}$$

$$T_{AC} = T_{AB} \frac{\sin 2^\circ}{\cos 30^\circ}$$

$$T_{AC} = 17,88 \frac{0,0349}{0,866}$$

$$T_{AC} = 0,72 \text{ kN}$$

$$T_{AB} = \frac{P}{\left(\cos 2^\circ - \frac{\sin 2^\circ}{\cos 30^\circ} \sin 30^\circ \right)}$$

$$T_{AB} = \frac{17,5 \text{ kN}}{\left(0,999 - \frac{0,0349}{0,866} 0,5 \right)}$$

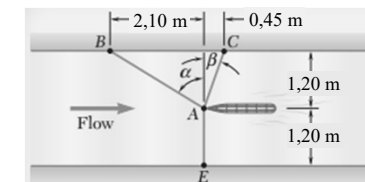
$$T_{AB} = 17,88 \text{ kN}$$

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Exemplo 5:

Como parte de um projeto de um novo veleiro deseja-se determinar a força de arraste a uma dada velocidade. Um modelo do casco é colocado em um canal para testes, sendo mantido alinhado com o eixo do canal por meio de três cabos presos a sua proa. Leituras de dinamômetros indicam que, para uma dada velocidade da água, a tração no cabo AB é de 200 N e no cabo AE de 300 N. Determine a força de arrasto no casco e a tração no cabo AC.



Repostas:

$$F_{\text{arrasto}} = 98,37 \text{ N}$$

$$T_{AC} = 214,5 \text{ N}$$

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Livro 1: Estática – Mecânica para engenharia (Hibbeler)

Capítulo 3, itens 3.1, 3.2, 3.3

Livro 2: Mecânica Vetorial para engenheiros - Estática (Beer e Johnston)

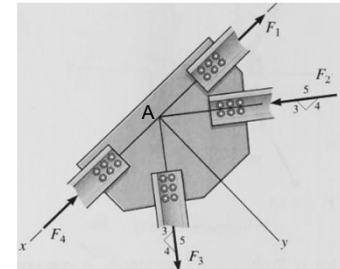
Capítulo 2, itens 2.8, 2.9, 2.11 e 2.11

Livro 3: Mecânica Estática (Merriam e Kraige)

Capítulo 2, Itens 2.3, 2.4, 2.5 e 2.6

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Determine as componentes x e y de cada força que atua sobre a chapa de ligação da estrutura tipo treliça que sustenta uma ponte. Obtenha a resultante das forças no ponto A.



Dados:

$$F_1 = 200 \text{ N}$$

$$F_2 = 400 \text{ N}$$

$$F_3 = 300 \text{ N}$$

$$F_4 = 300 \text{ N}$$